DO YOU KNOW?

(i) If X_1 and X_2 are random variables and C is a constant, then $CX_1, X_1 + X_2, X_1X_2, X_1 - X_2$ are also random variables.

(ii) If X is a random variable, then $(i)\frac{1}{X}$ and $(ii)|X|$ are also random variables.

Types of Random Variable:

Random variables are classified into two types namely discrete and continuous random variables. These are important for practical applications in the field of Mathematics and Statistics. The above types of random variable are defined with examples as follows.

6.1.2 Discrete random variable

Definition 6.2

A variable which can assume finite number of possible values or an infinite sequence of countable real numbers is called a discrete random variable.

Examples of discrete random variable:

- Marks obtained in a test.
- Number of red marbles in a jar.
- Number of telephone calls at a particular time.
- Number of cars sold by a car dealer in one month, etc.,

For instance, three responsible persons say, P_1, P_2 and P_3 are asked about their opinion in favour of building a model school in a certain district. Each person's response is recorded as Yes (Y) or No (N). Determine the random variable that could be of interest in this regard. The possibilities of the response are as follows

Table 6.1

Possibilities	P_1	P_2	P_3	Values of Random variable (Number of Yes those who are given)
1.	Y	Y	Y	3
2.	Y	Y	N	2
3.	Y	N	Y	2
4.	Y	N	N	1
5.	N	Y	Y	2
6.	N	Y	N	1
7.	N	N	Y	1
8.	N	N	N	0

Form the above table, the discrete random variable take values 0, 1, 2 and 3.

Probability Mass function

Definition 6.3

If X is a discrete random variable with distinct values $x_1, x_2, ..., x_n, ...$, then the function, denoted by $P_X(x)$ and defined by

$$P_X(x) = \mathrm{p(x)} = \begin{cases} \mathrm{P(X = x_i)} = p_i = p(x_i) & \text{if } x = x_i, i = 1, 2, ..., n, ... \\ 0 & \text{if } x \neq x_i \end{cases}$$

This is defined to be the probability mass function or discrete probability function of X.

The probability mass function $p(x)$ must satisfy the following conditions

(i) $p(x_i) \geq 0 \; \forall \, i$, (ii) $\sum_{i=1}^{\infty} p(x_i) = 1$

Example 6.1

The number of cars in a household is given below.

No. of cars	0	1	2	3	4
No. of Household	30	320	380	190	80

Estimate the probability mass function. Verify $p(x_i)$ is a probability mass function.

Solution:

Let X be the number of cars

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